Efficacy and Safety of Cardimexa in Heart failure with reduced ejection fraction: A Phase 3 Randomised, Double-Blind, Placebo-Controlled Trial

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Note: Entirely fictional study with invented drug and data, created for training. Do not use for clinical decisions.

Abstract

Background: Cardimexa (CDX) is a twice-daily oral neprilysin-pathway modulator under investigation for heart failure with reduced ejection fraction. **Methods:** In this multicentre trial, 1820 adults were randomised 1:1 to Cardimexa or placebo for 52 weeks, stratified by baseline severity and region. The primary endpoint was composite of cardiovascular death or HF hospitalisation. **Results:** The primary endpoint was 0.79 with Cardimexa versus 1.0 with placebo (p 0.003). Symptomatic hypotension occurred in 11.4% of Cardimexa patients and 7.1% of placebo patients. **Conclusions:** Cardimexa met its primary endpoint with a manageable safety profile in heart failure with reduced ejection fraction.

Methods

Eligible participants were aged 18 to 75 years with a confirmed diagnosis of heart failure with reduced ejection fraction and provided written informed consent. Exclusion criteria included pregnancy, severe hepatic impairment (Child-Pugh C), and concurrent use of strong CYP3A4 inducers. Randomisation used an interactive web-response system. The analysis followed the intention-to-treat principle with multiple imputation for missing data.

Results

Characteristic	Cardimexa	Placebo
Participants (n)	910	910
Mean age (years)	54.1	61.3
Female (%)	56	54
Primary endpoint	0.79	1.0
p-value	0.003	-
Symptomatic hypotension (%)	11.4	7.1
Discontinuation due to AE (%)	6.7	3.3

Serious adverse events were reported in 11.7% of the Cardimexa group and 9.0% of the placebo group. There were 6 deaths in the Cardimexa arm and 3 in the placebo arm, none judged related to study treatment.

Discussion

The effect size observed with Cardimexa is clinically meaningful. Limitations include the 52-week duration, under-representation of patients aged over 75 years, and limited ethnic diversity. Long-term safety beyond one year requires an open-label extension, which is ongoing.

References

1. Synthetic reference list omitted. 2. Illustrative Guidelines on Management, 2025.